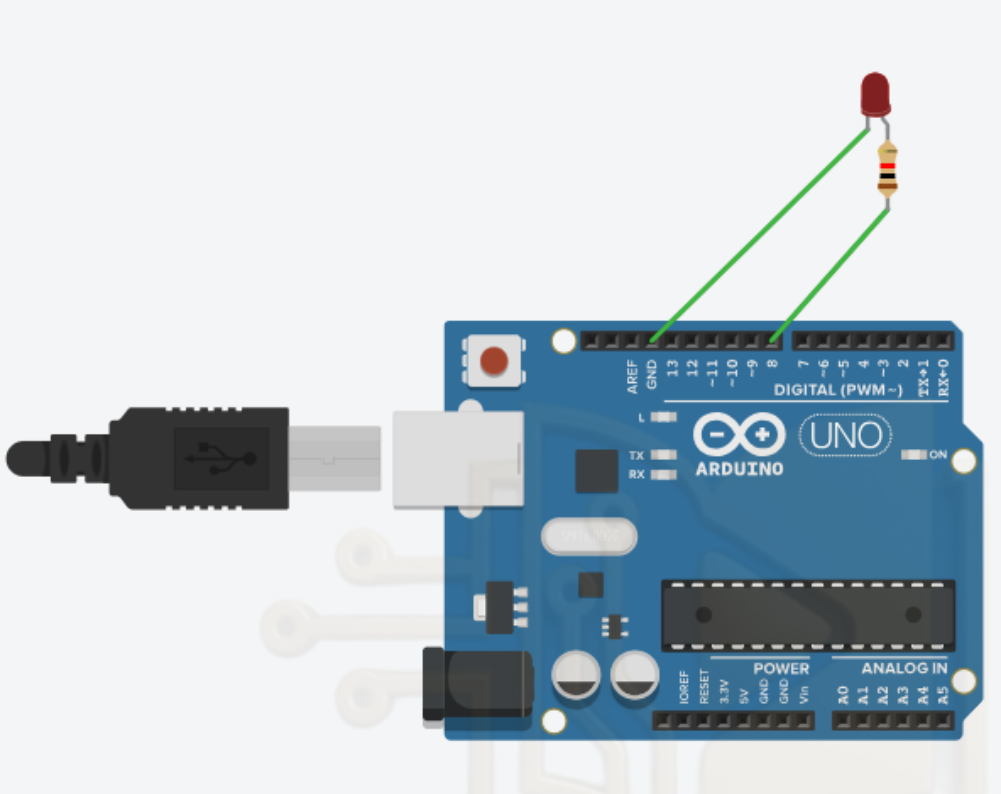
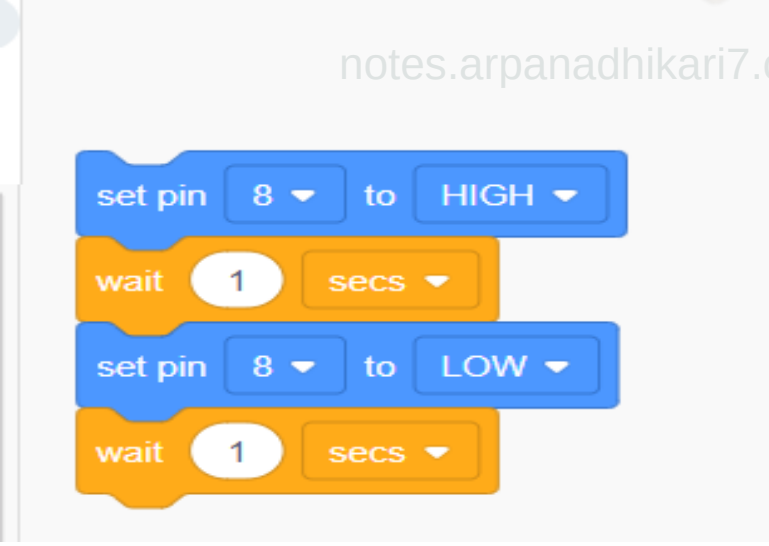
IoT projects:

1. Simple LED on and Off:

Circuit Configuration:

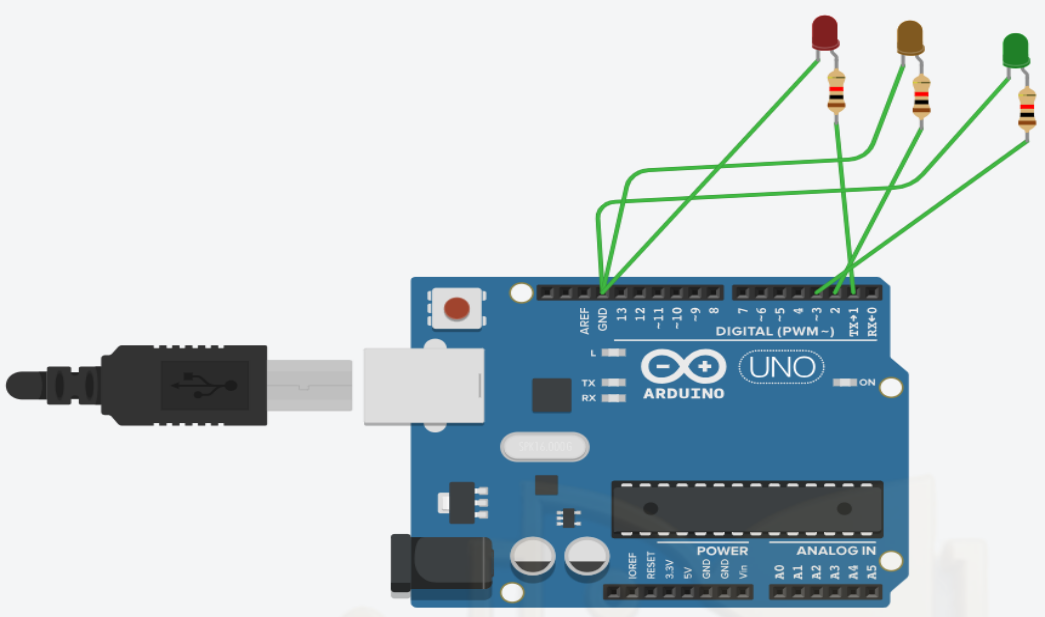


Code:

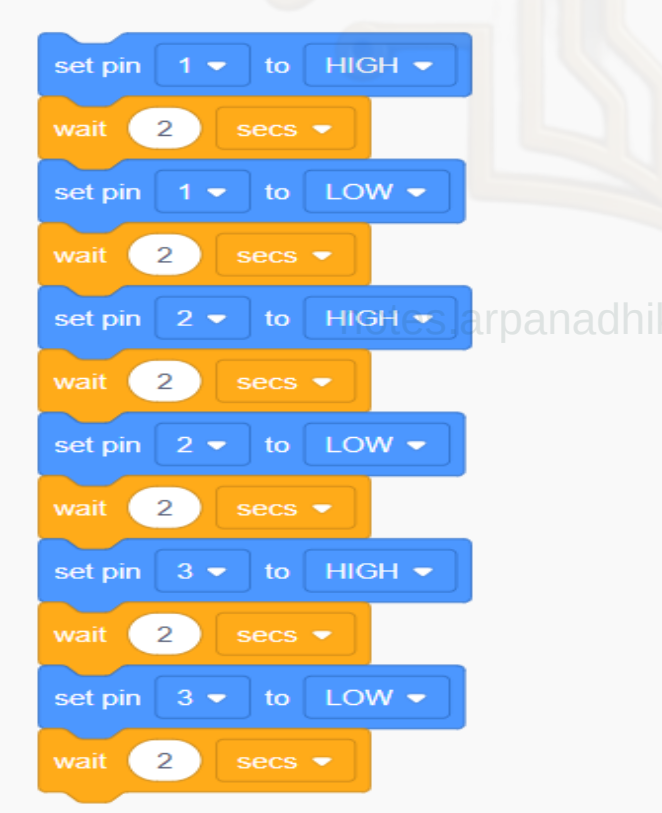


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2. 3 Led RGB with a delay of 2sec:

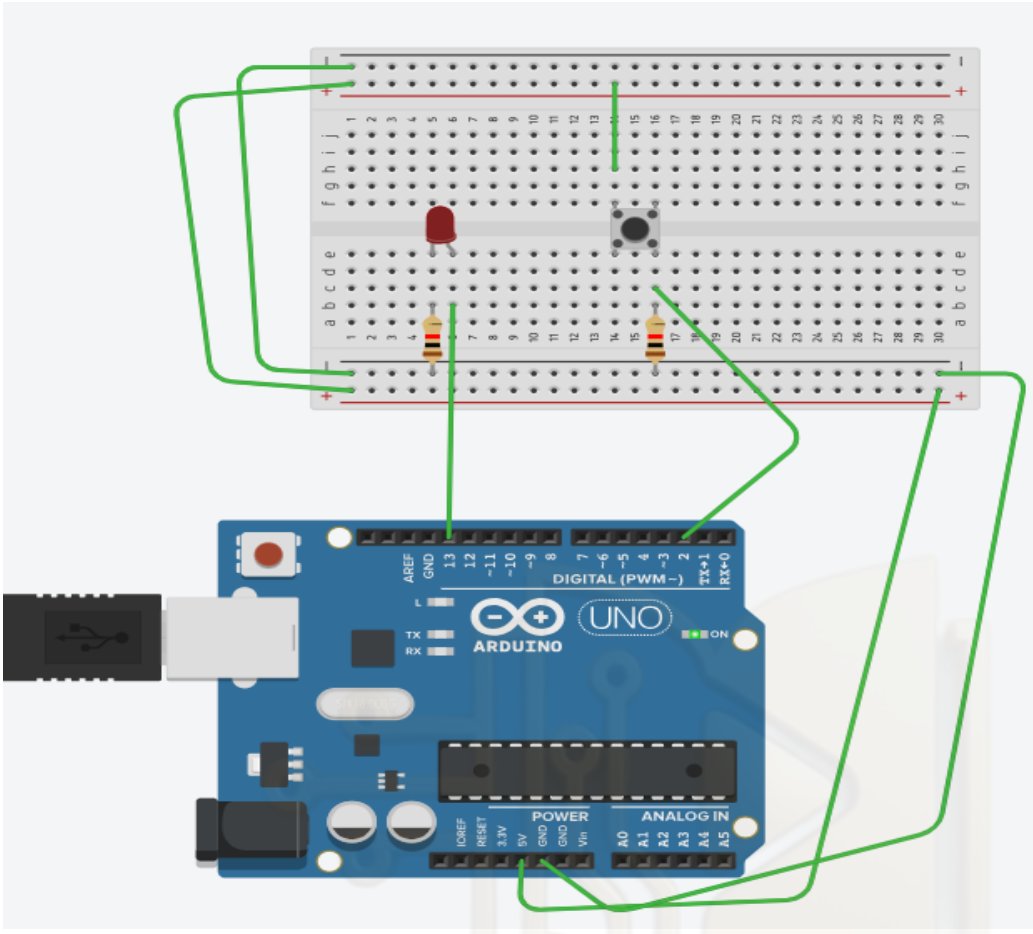


Code:

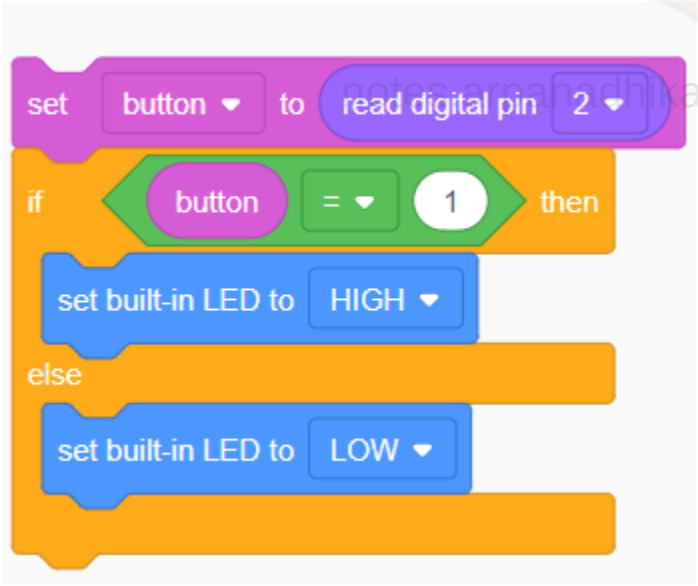


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3. Led On and Off with push button:

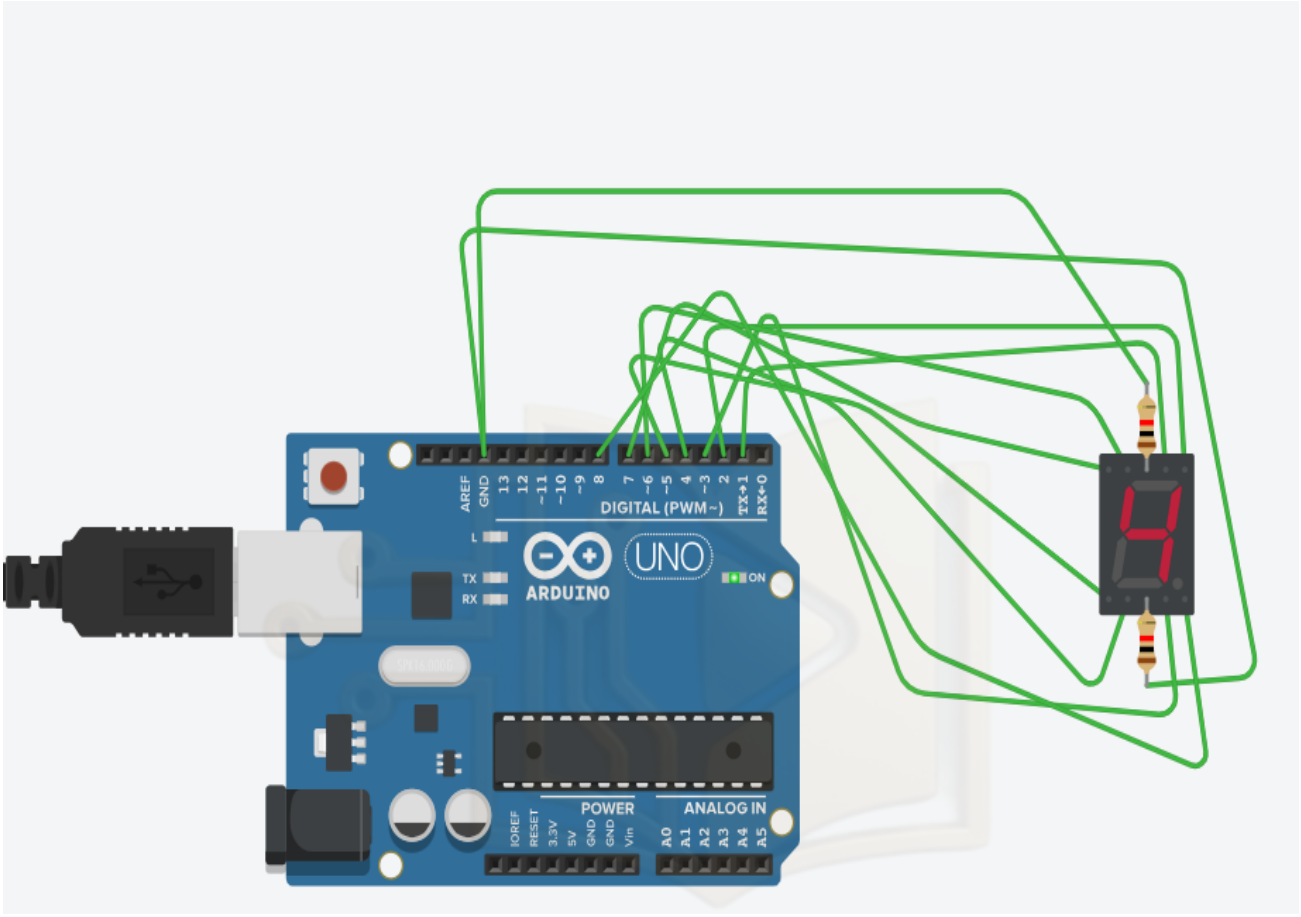


Code:



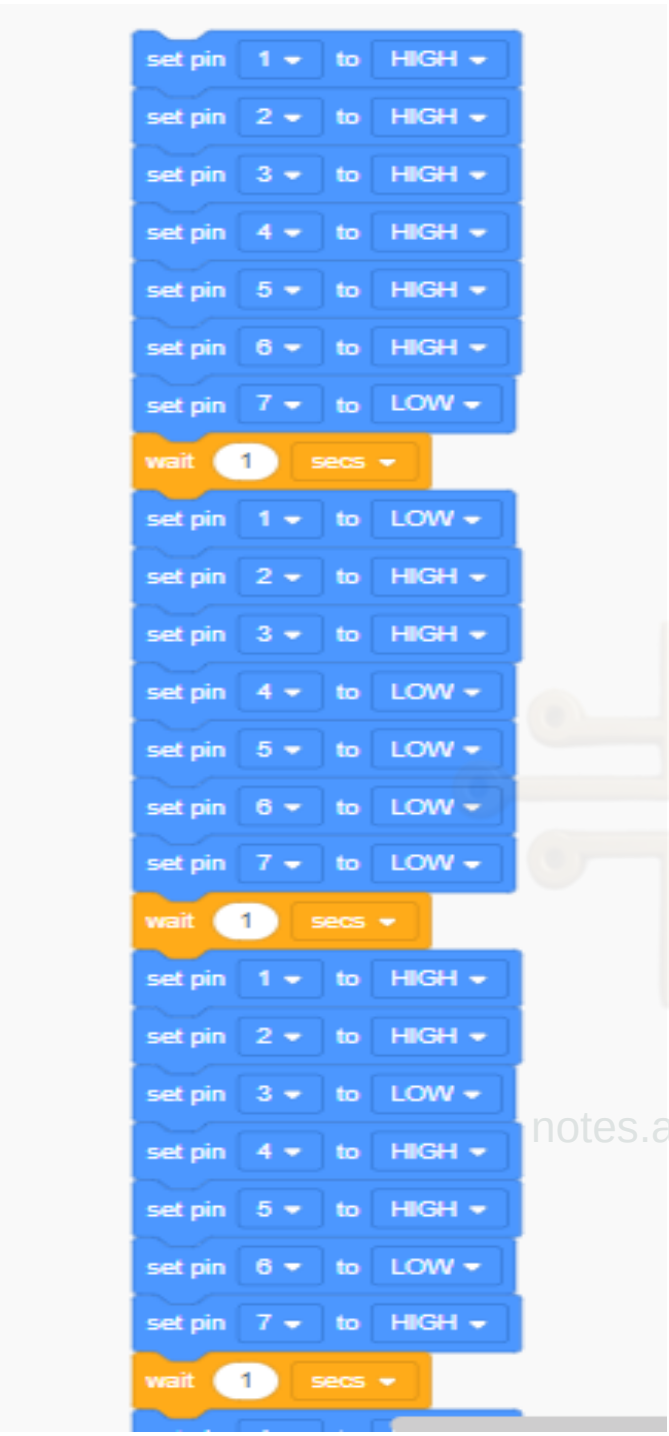
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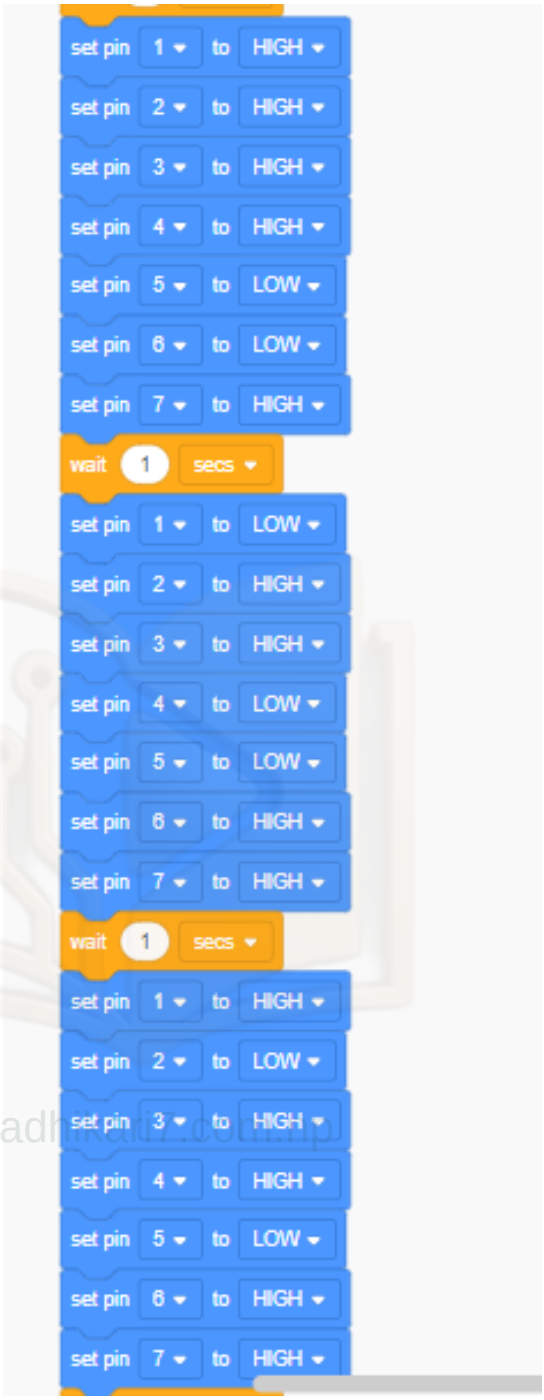
4. 7-Segment Display:



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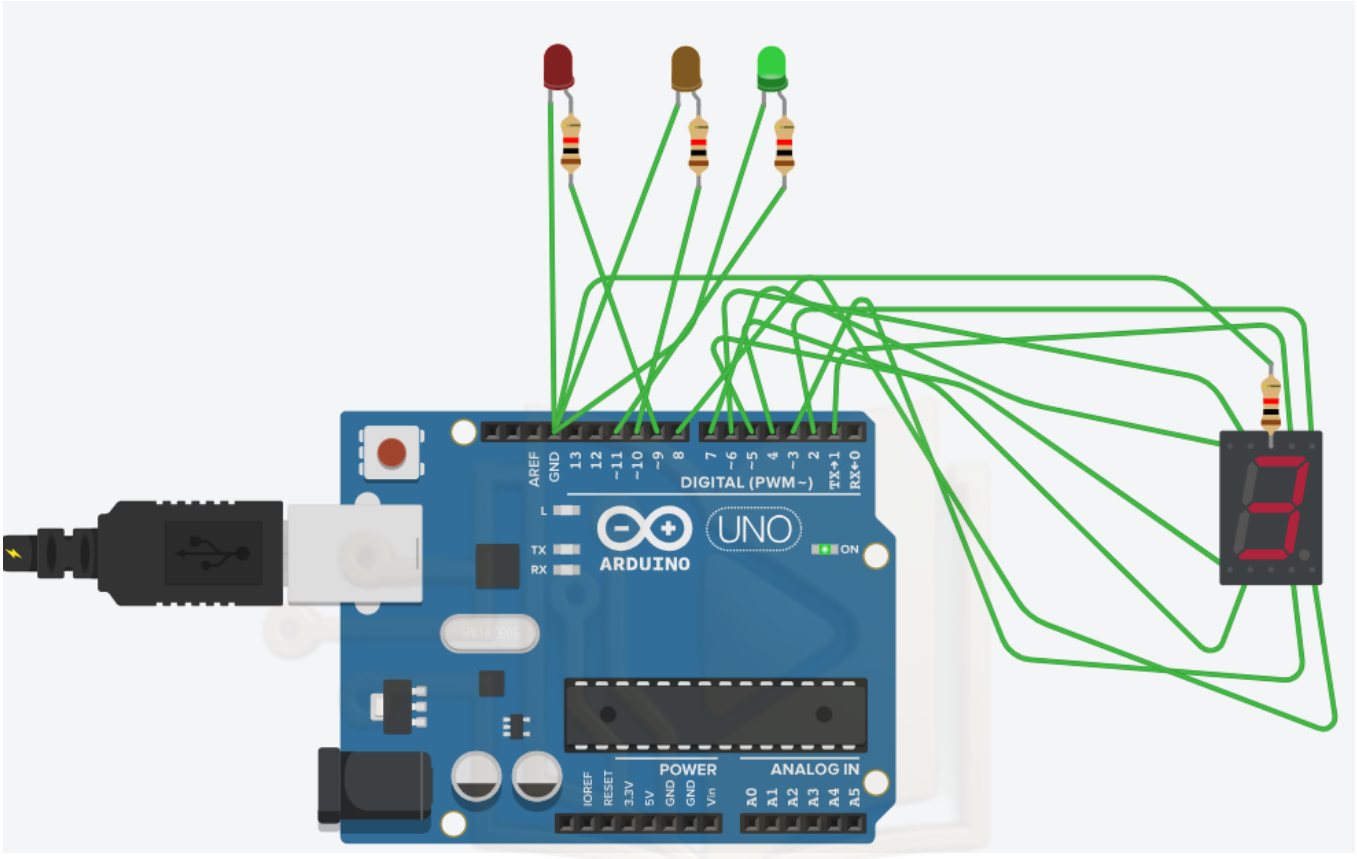
Code:





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5. Seven-segment display with LED traffic light:



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Code:

```
const int redLED    = 9;
const int yellowLED = 10;
const int greenLED  = 11;
const int segPins[7] = {1, 2, 3, 4, 5, 6, 7};
const byte digits[10][7] = {
  {1,1,1,1,1,1,0}, // 0
  {0,1,1,0,0,0,0}, // 1
  {1,1,0,1,1,0,1}, // 2
  {1,1,1,1,0,0,1}, // 3
  {0,1,1,0,0,1,1}, // 4
  {1,0,1,1,0,1,1}, // 5
  {1,0,1,1,1,1,1}, // 6
  {1,1,1,0,0,0,0}, // 7
  {1,1,1,1,1,1,1}, // 8
  {1,1,1,1,0,1,1}  // 9
};
void displayDigit(int num) {
  for (int i = 0; i < 7; i++) {
    digitalWrite(segPins[i], digits[num][i]);
  }
}
void setup() {
  pinMode(redLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(greenLED, OUTPUT);

  for (int i = 0; i < 7; i++) {
    pinMode(segPins[i], OUTPUT);
  }
}

void loop() {
  // GREEN light countdown (5 → 0)
  digitalWrite(greenLED, HIGH);
  for (int i = 5; i >= 0; i--) {
    displayDigit(i);
    delay(1000);
  }
  digitalWrite(greenLED, LOW);

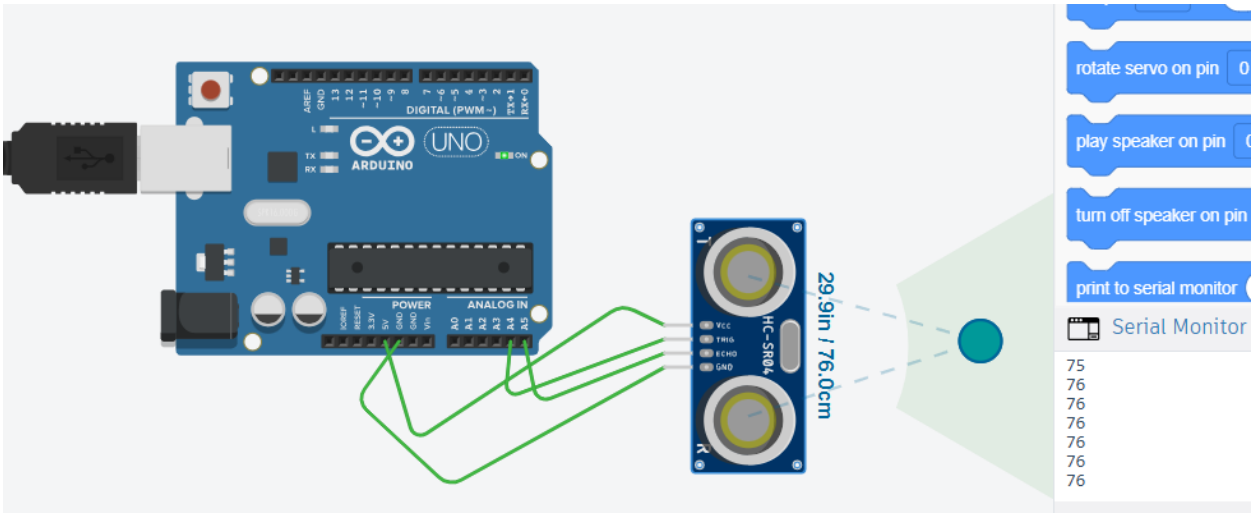
  // YELLOW light countdown (2 → 0)
  digitalWrite(yellowLED, HIGH);
  for (int i = 2; i >= 0; i--) {
    displayDigit(i);
    delay(1000);
  }
  digitalWrite(yellowLED, LOW);

  // RED light countdown (5 → 0)
  digitalWrite(redLED, HIGH);
  for (int i = 5; i >= 0; i--) {
    displayDigit(i);
    delay(1000);
  }
  digitalWrite(redLED, LOW);
}
```

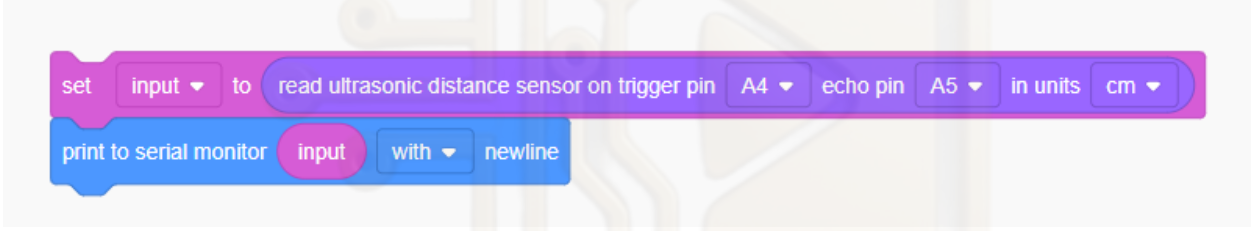
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6. Ultrasonic Sensor Interfacing:

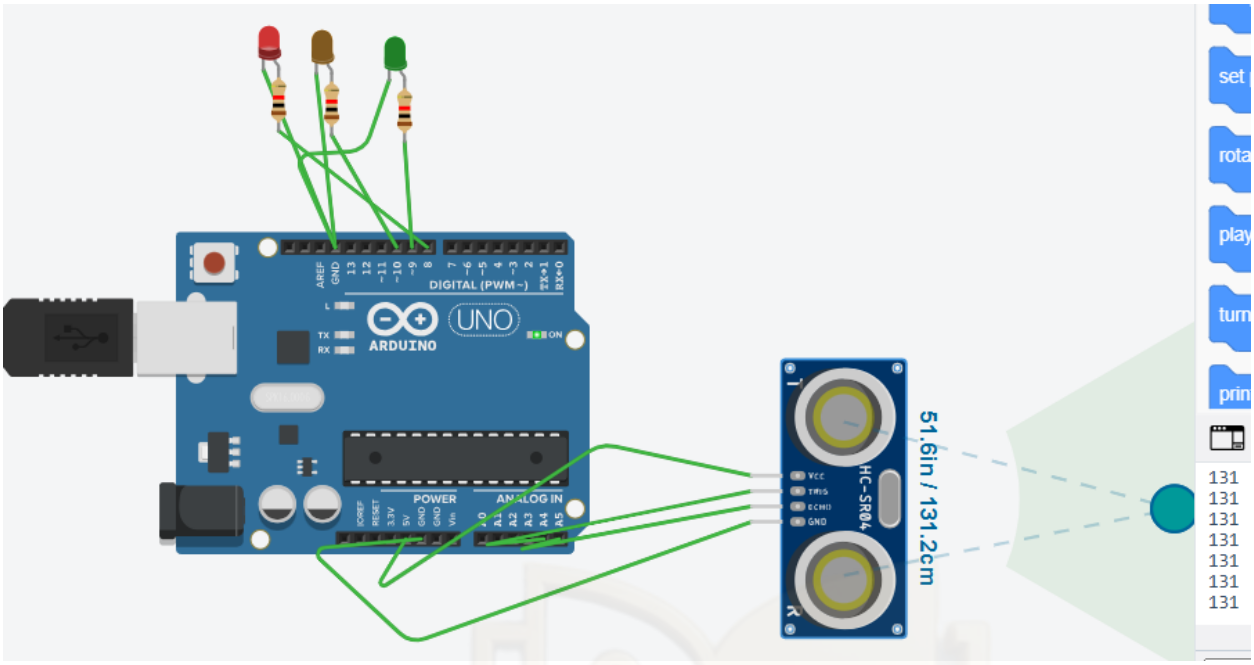


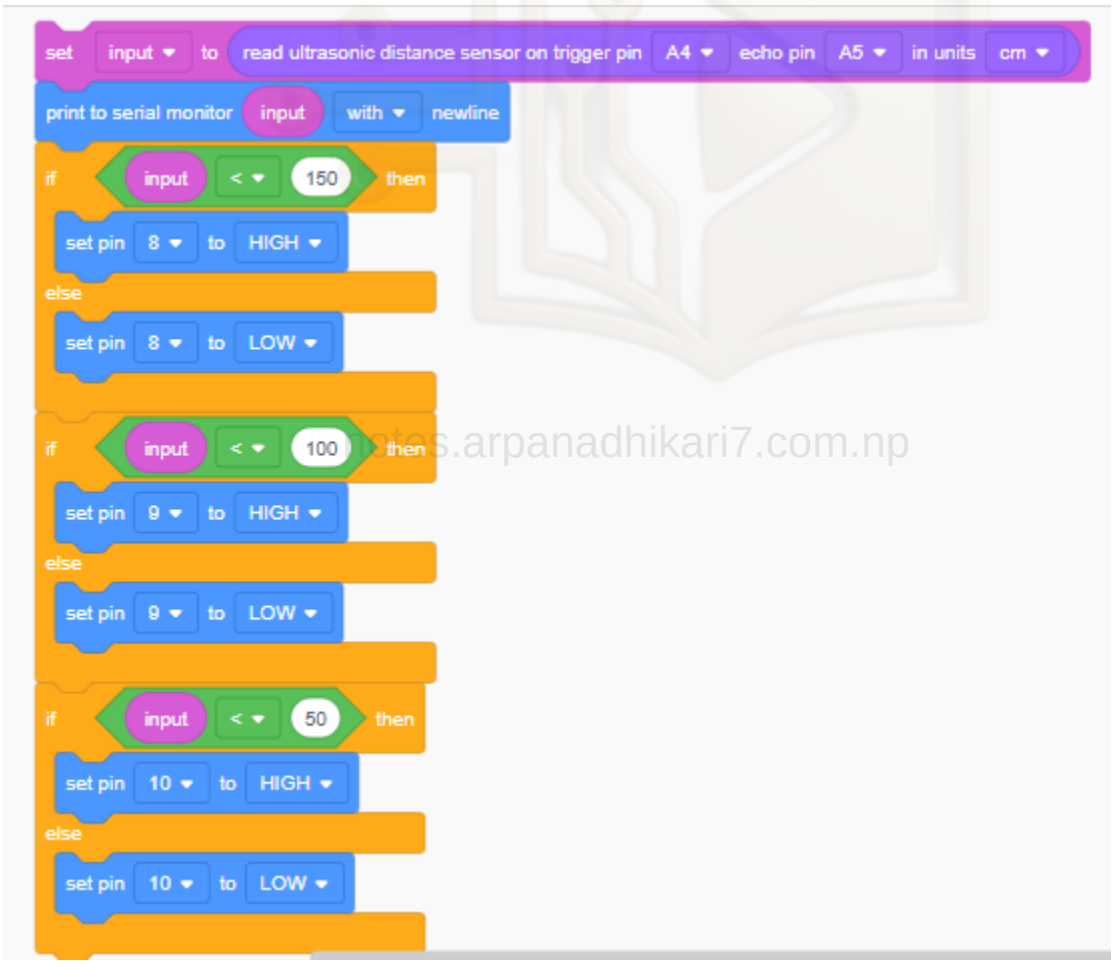
Code:



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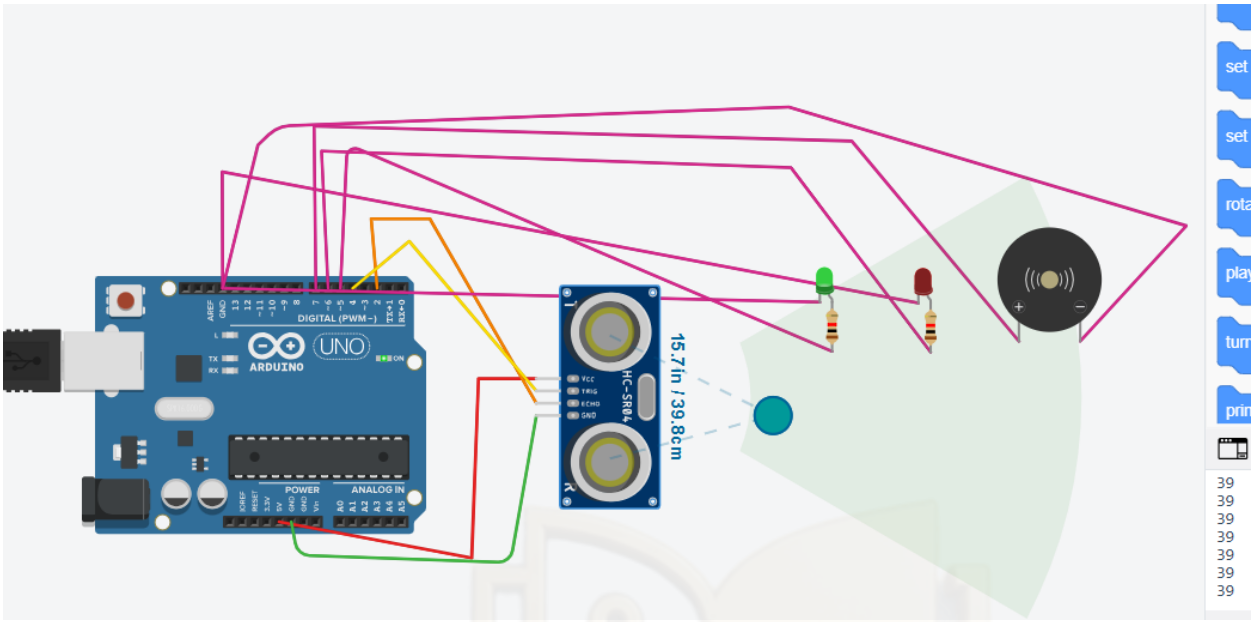
7. Distance Measurement with LED & Arduino and Ultrasonic sensors:



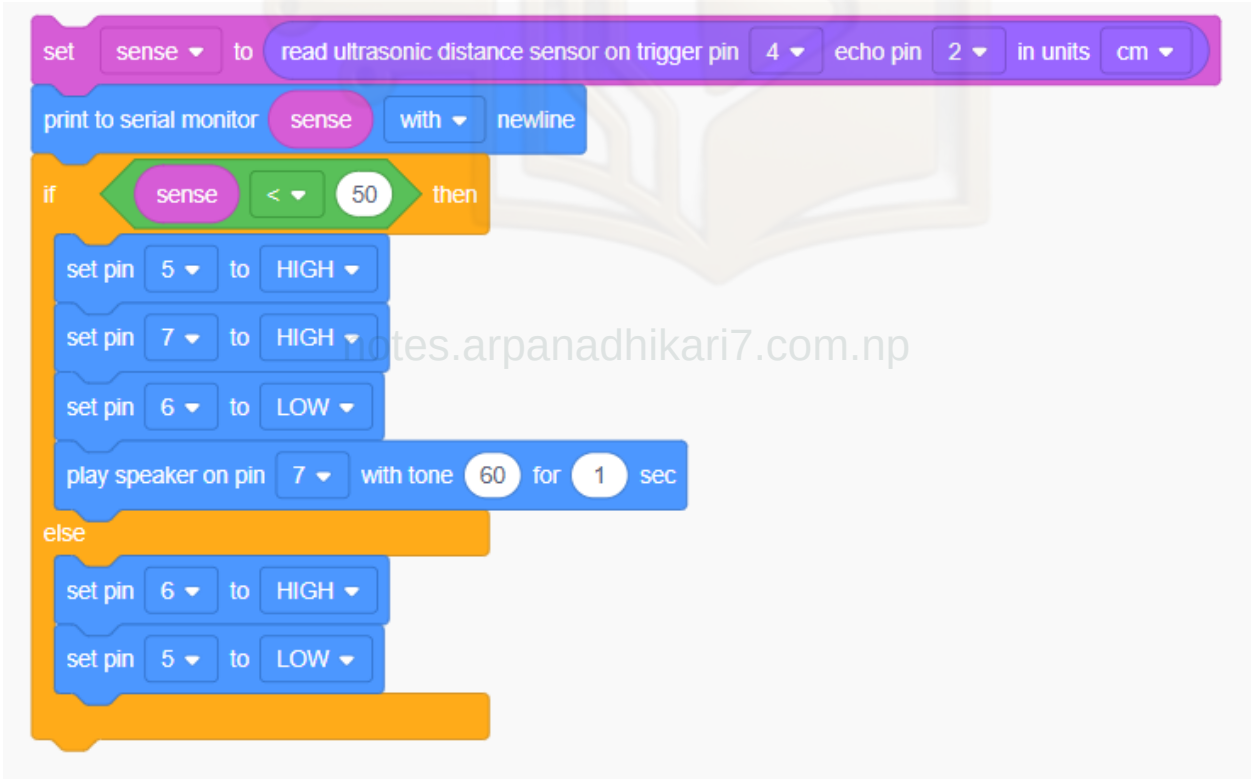


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8. Distance Measurement with LED & Arduino and Ultrasonic sensors with an alarm as well:

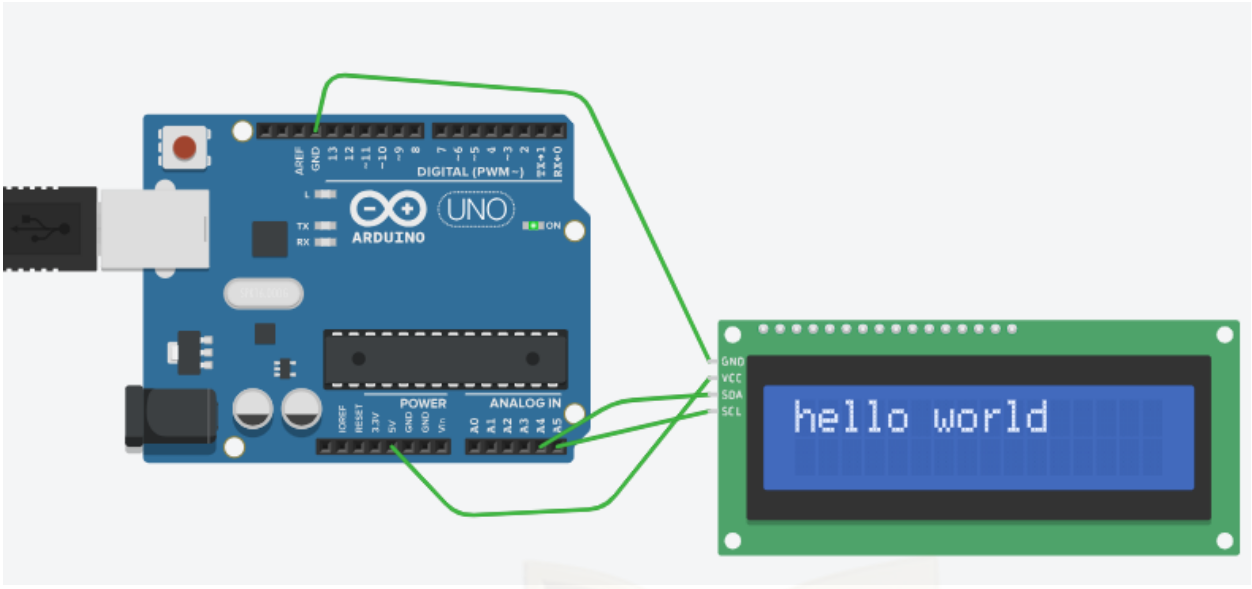


Code:

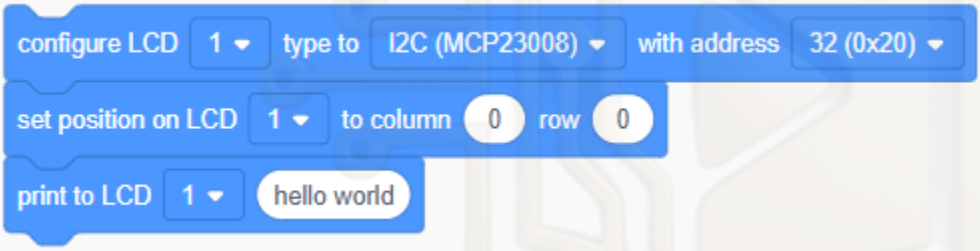


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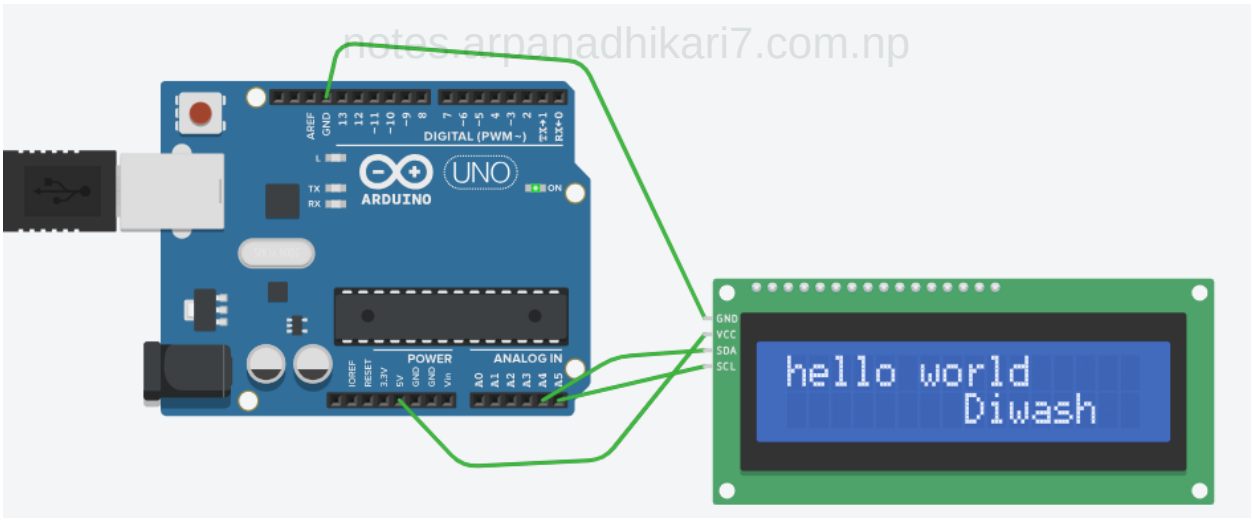
9. Print Hello world



Code:

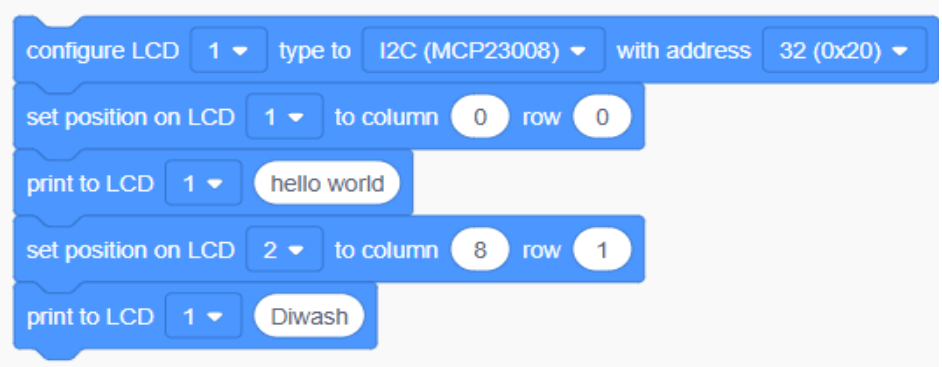


Print name:

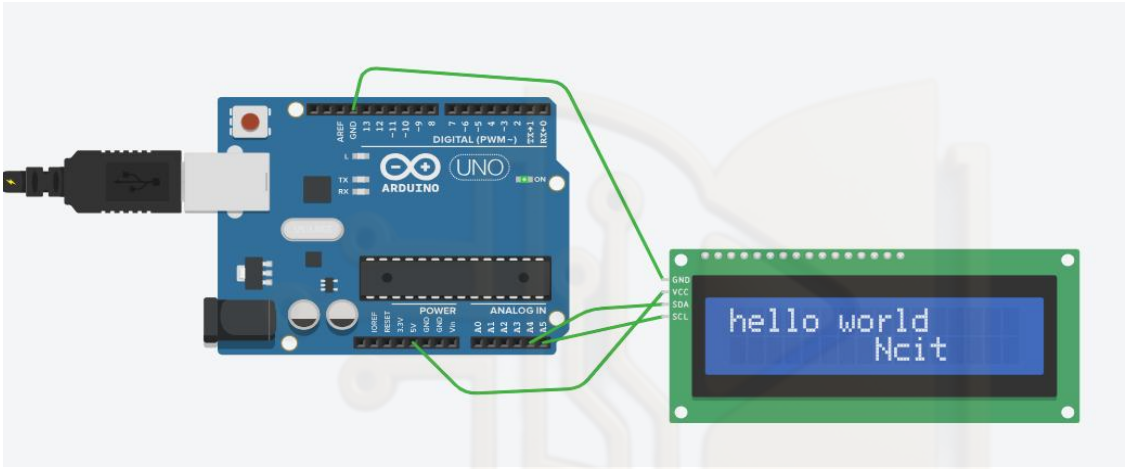


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Code:



Print ncit:

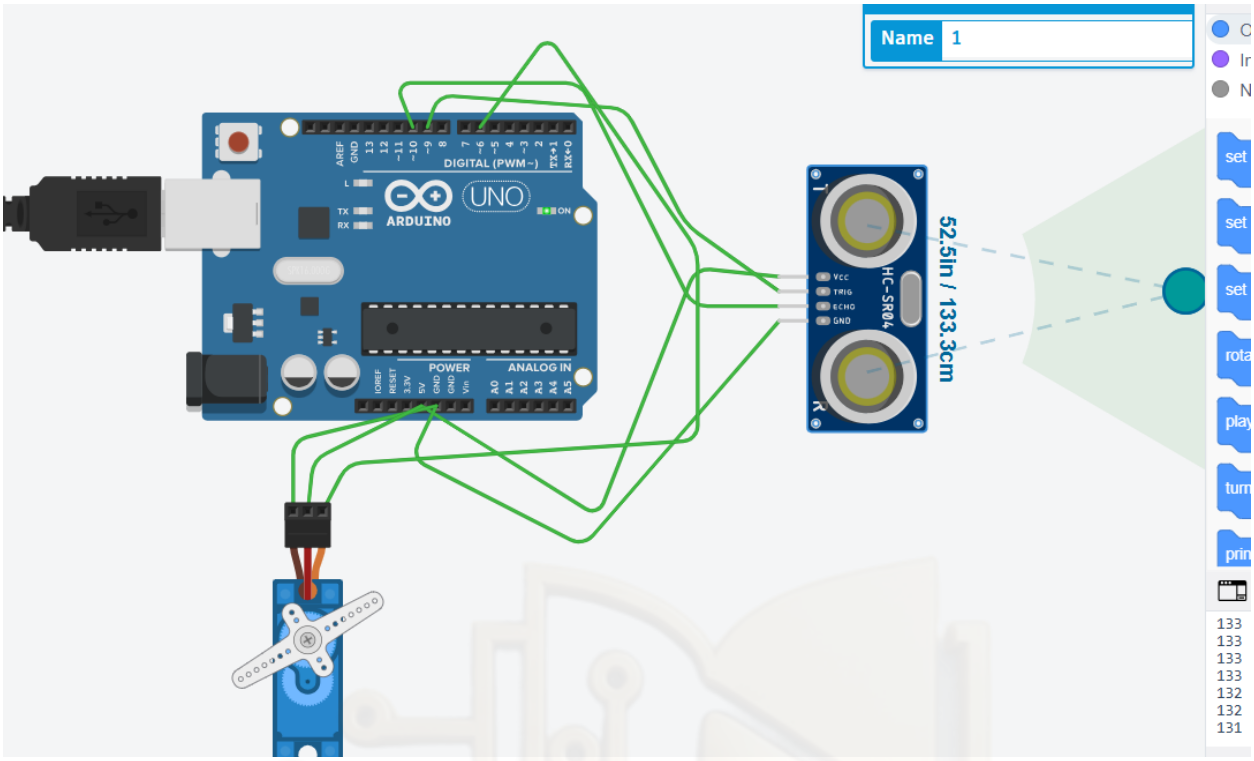


Code:

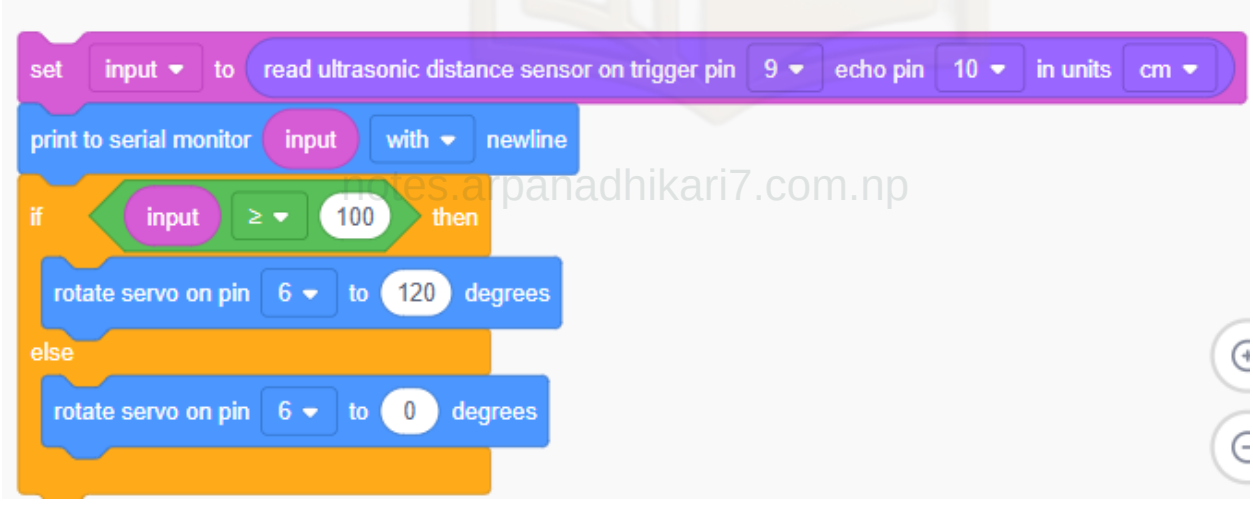


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10. Ultrasonic sensor with servo motor:

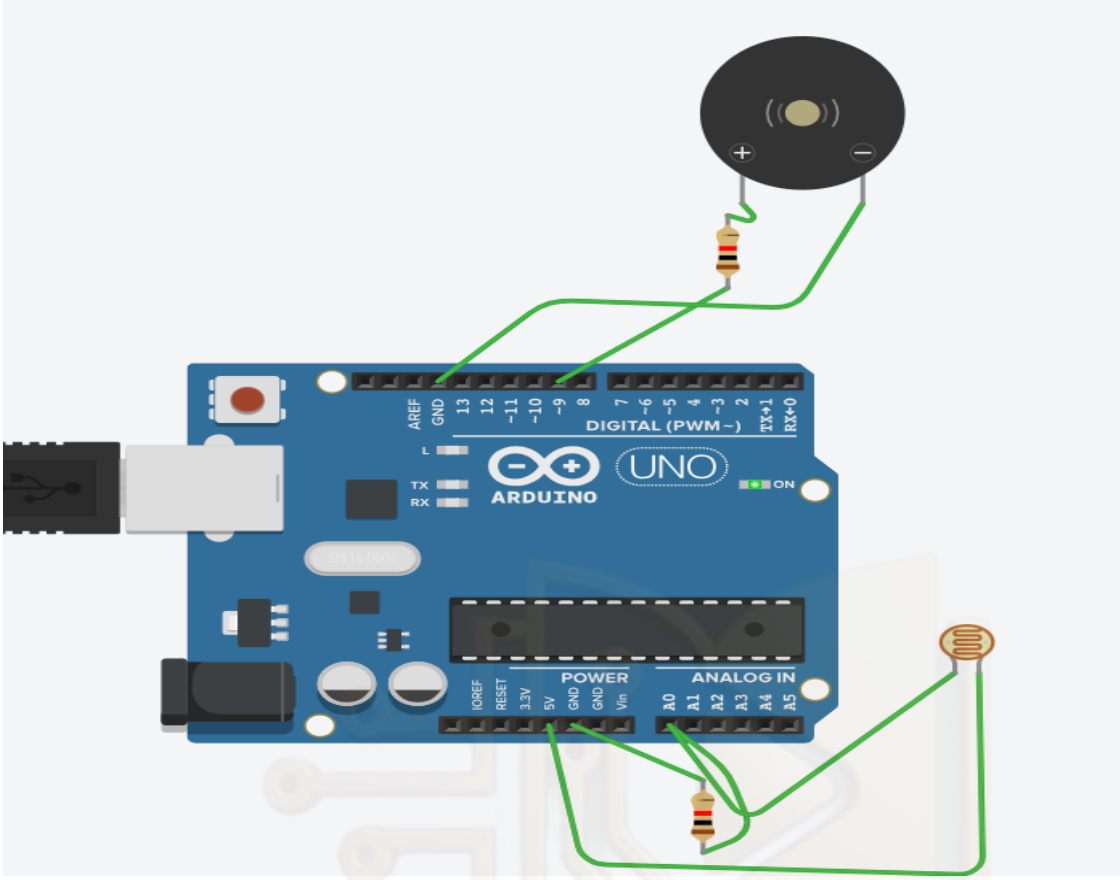


Code:



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11. Tone Pitch follower:

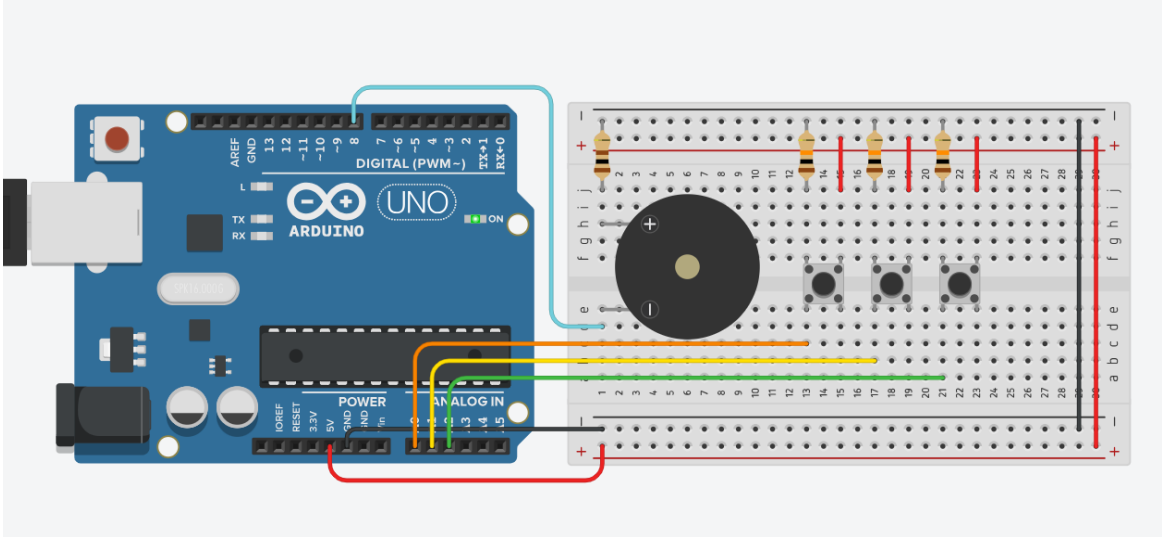


Code:



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12. Tone keyboard:

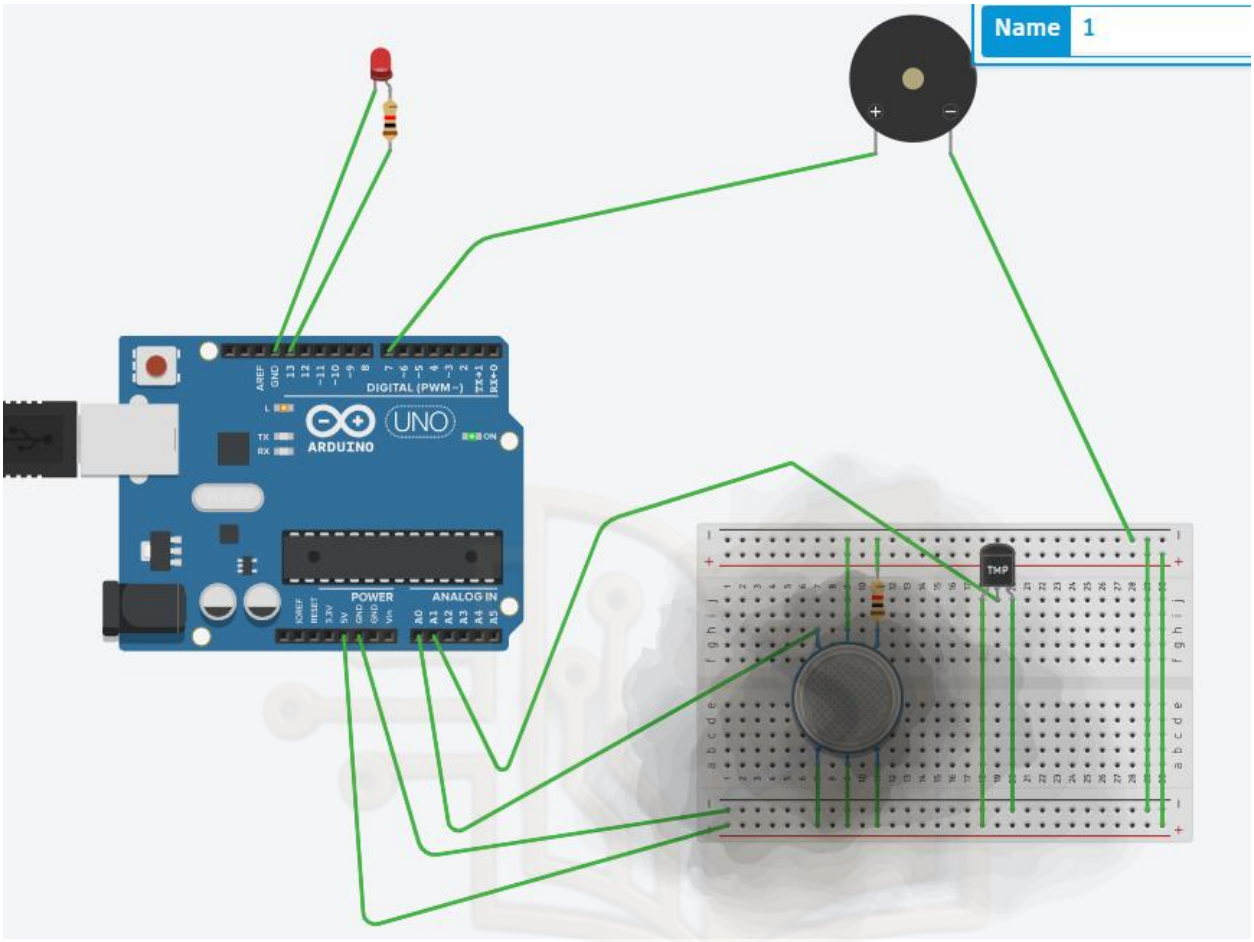


Code:



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13. Interfacing Temperature Sensor with Arduino:



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Text

1 (Arduino Uno R3

```
const int gasPin = A0;
const int tempPin = A1;
const int buzzer = 7;
const int led = 13;
int gasThreshold = 300;
float tempThreshold = 50.0;
void setup() {
  pinMode(buzzer, OUTPUT);
  pinMode(led, OUTPUT);
  Serial.begin(9600);
  Serial.println("Fire Alarm System Ready");
}
void loop() {
  int gasValue = analogRead(gasPin);
  int tempValue = analogRead(tempPin);
  float voltage = tempValue * (5.0 / 1023.0);
  float tempC = (voltage - 0.5) * 100.0;
  Serial.print("Gas Value = ");
  Serial.print(gasValue);
  Serial.print(" | Temp = ");
  Serial.print(tempC);
  Serial.println(" °C");
  if (gasValue > gasThreshold || tempC > tempThreshold) {
    digitalWrite(led, HIGH);
    digitalWrite(buzzer, HIGH);
    delay(200);
    digitalWrite(buzzer, LOW);
    delay(200);
  } else {
    digitalWrite(led, LOW);
    digitalWrite(buzzer, LOW);}
  delay(500);}
```

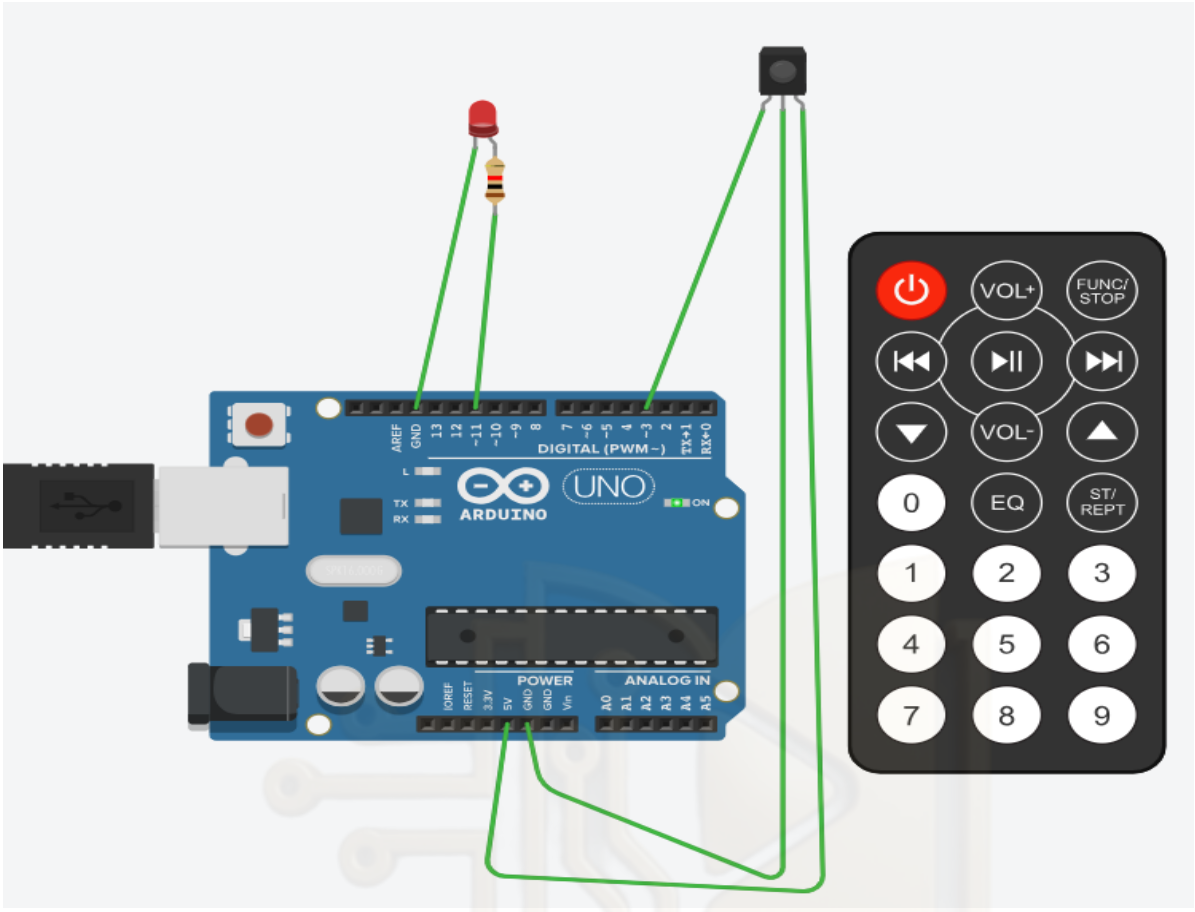
Serial Monitor

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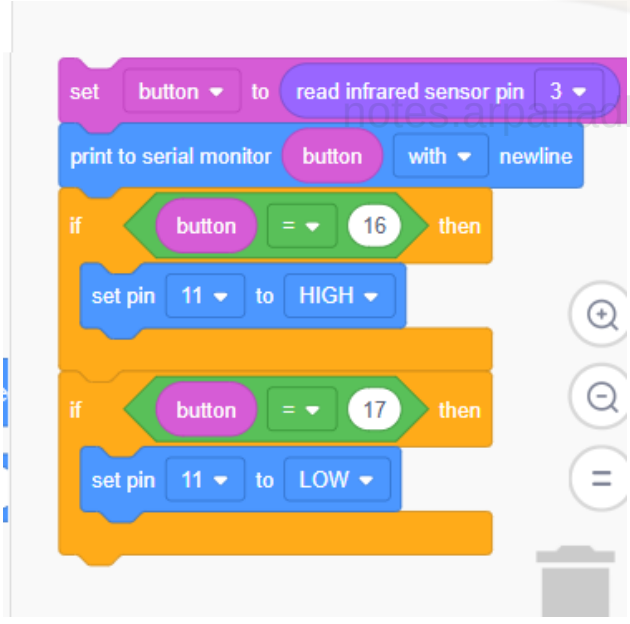
```
Gas Value = 307 | Temp = 24.78 Â°C
Gas Value = 307 | Temp = 24.78 Â°C
Gas Value = 307 | Temp = 24.78 Â°C
Gas Value = 307 | Temp = 24.78 Â°C
Gas Value = 307 | Temp = 24.78 Â°C
Gas Value = 307 | Temp = 24.78 Â°C
Gas Value = 307 | Temp = 24.78 Â°C
```

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14. Controlling LEDs using an IR Remote:

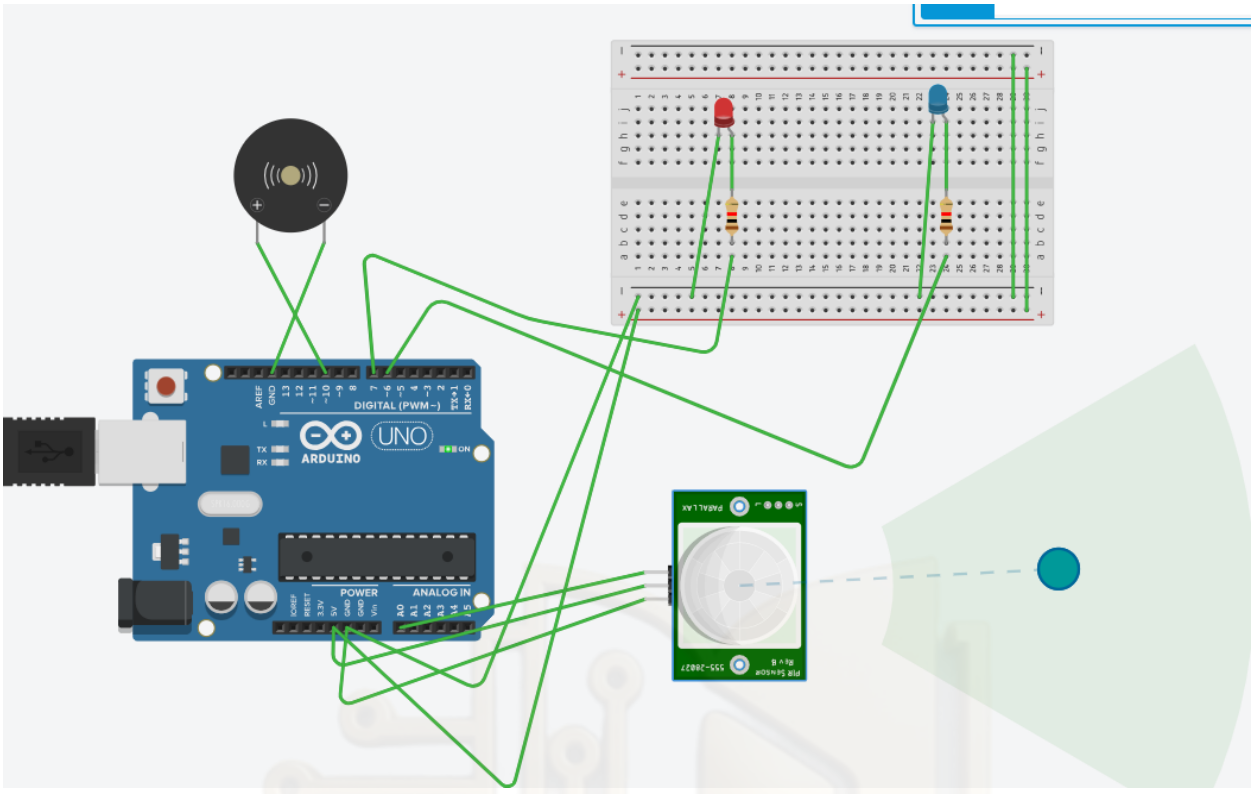


Code:

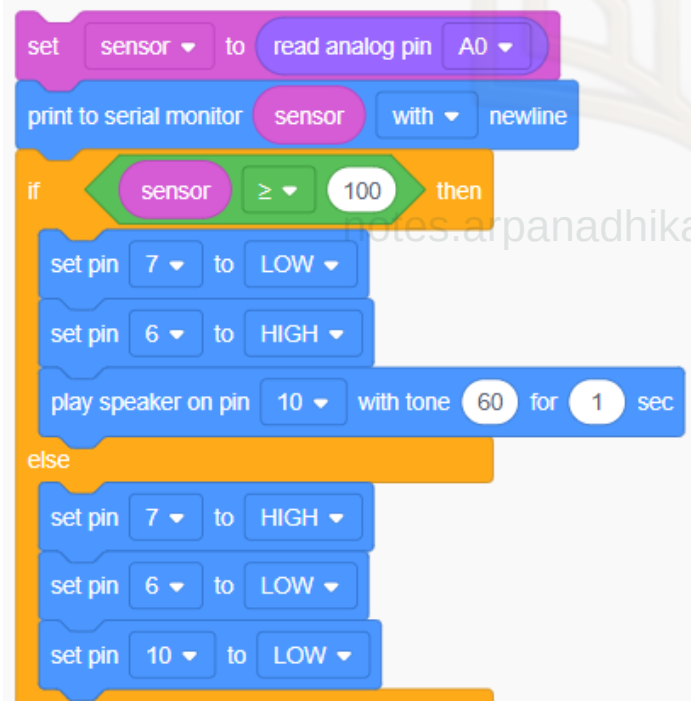


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15. PIR motion sensor:

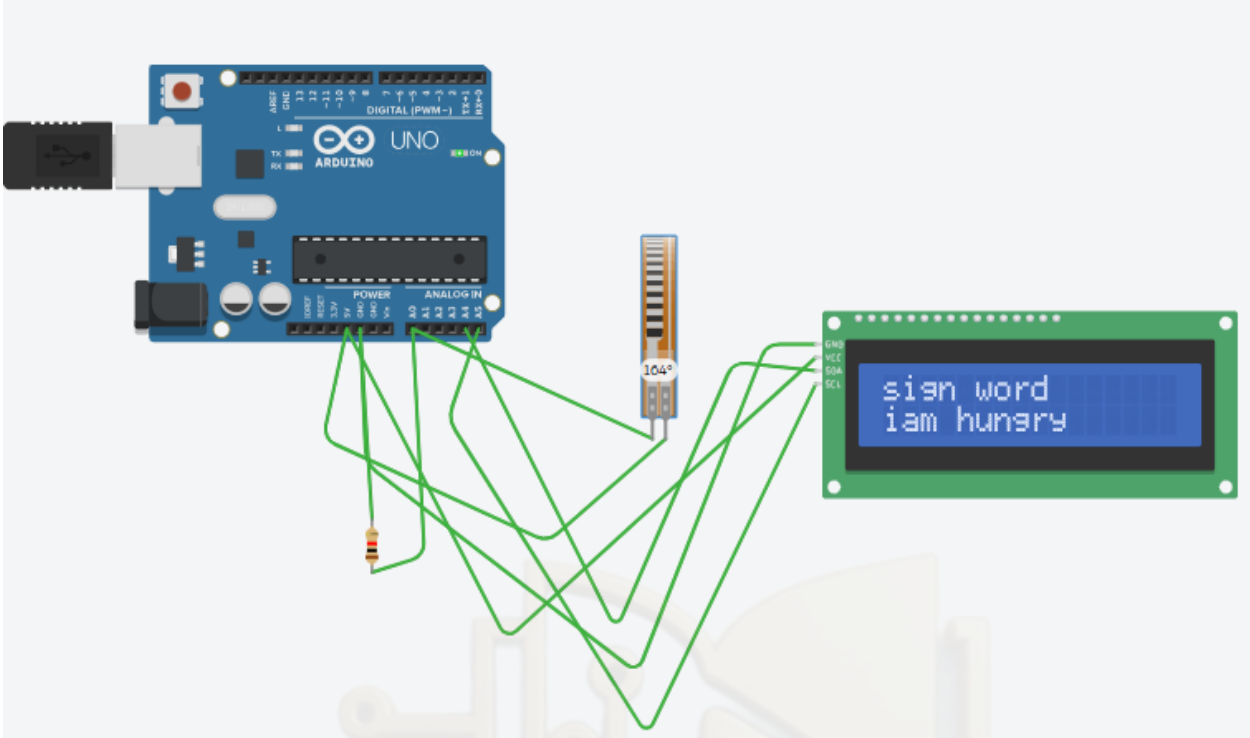


Code:



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16. Flex sensor for sign language:

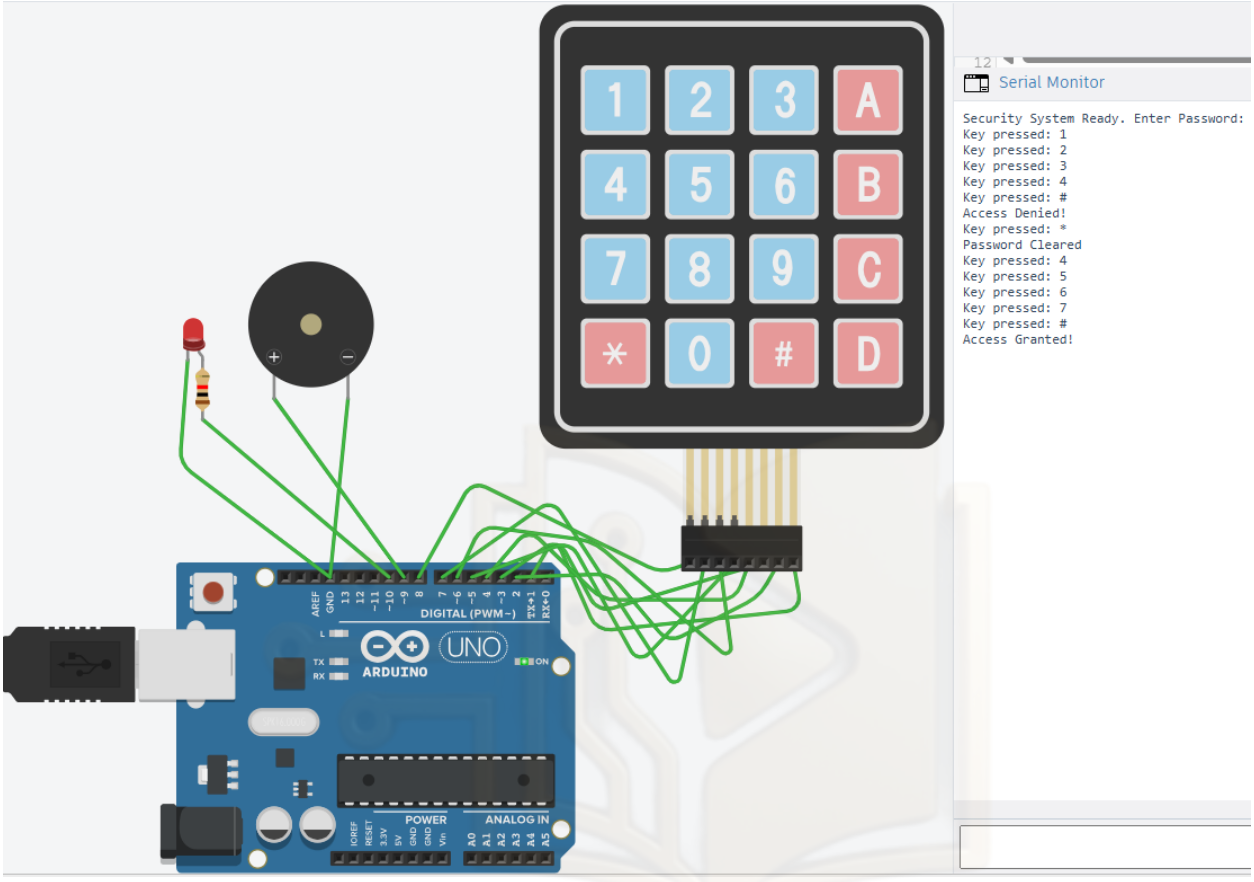


Code:



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17. Password-Protected Security System project using Arduino, Keypad, Piezo Buzzer, and LED.



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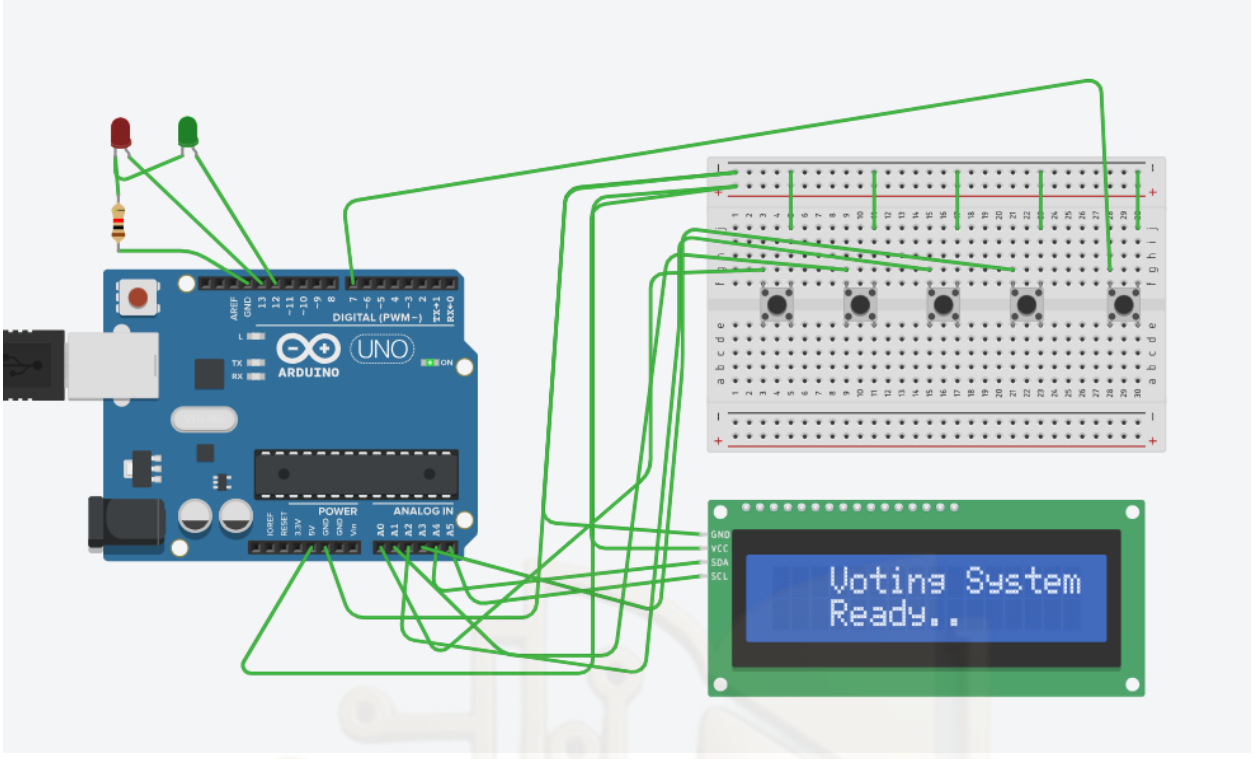
Code:

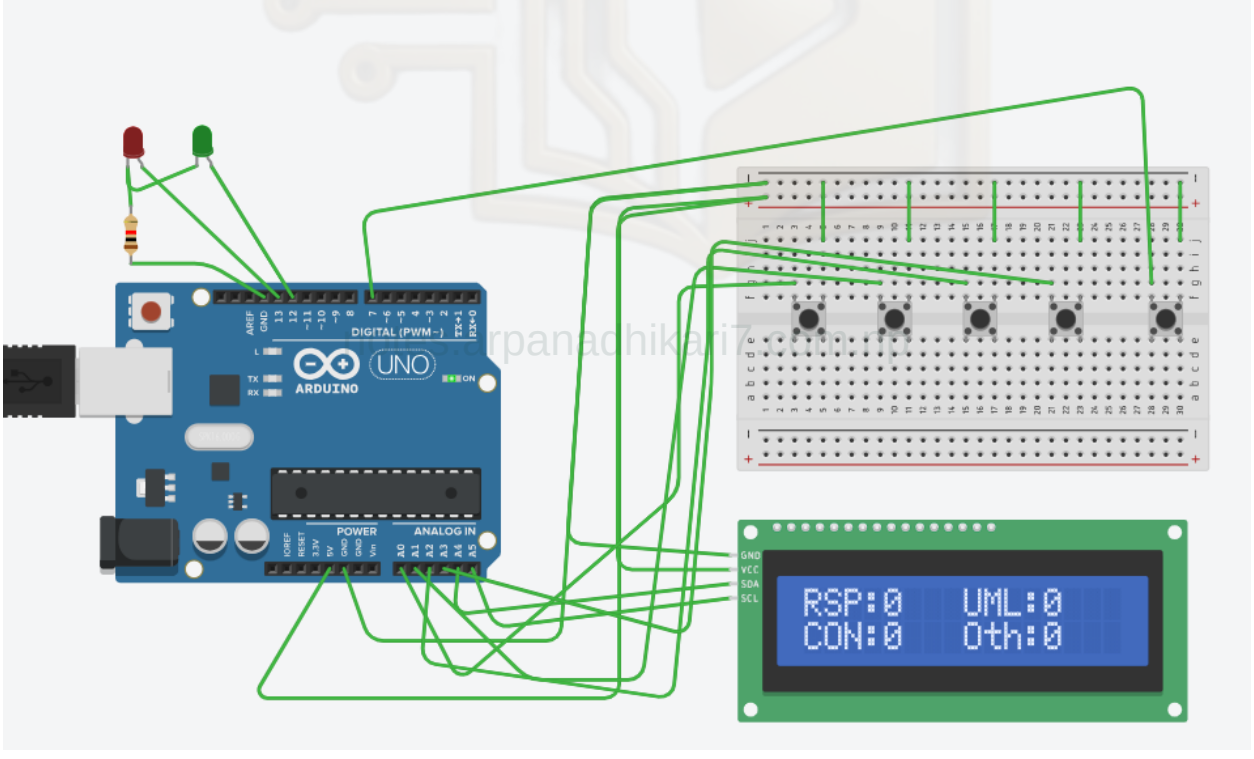
```
#include <Keypad.h>
const byte ROWS = 4;
const byte COLS = 4;
char keys[ROWS][COLS] = {
  {'1','2','3','A'},
  {'4','5','6','B'},
  {'7','8','9','C'},
  {'*','0','#','D'} };
byte rowPins[ROWS] = {8,7,6,5};
byte colPins[COLS] = {4,3,2,1};
Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS,
const int ledPin    = 10;
const int buzzerPin = 9;
String password      = "4567";
String inputPassword = "";
void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);
  Serial.begin(9600);
  Serial.println("Security System Ready. Enter Password:");
}
void loop() {
  char key = keypad.getKey();

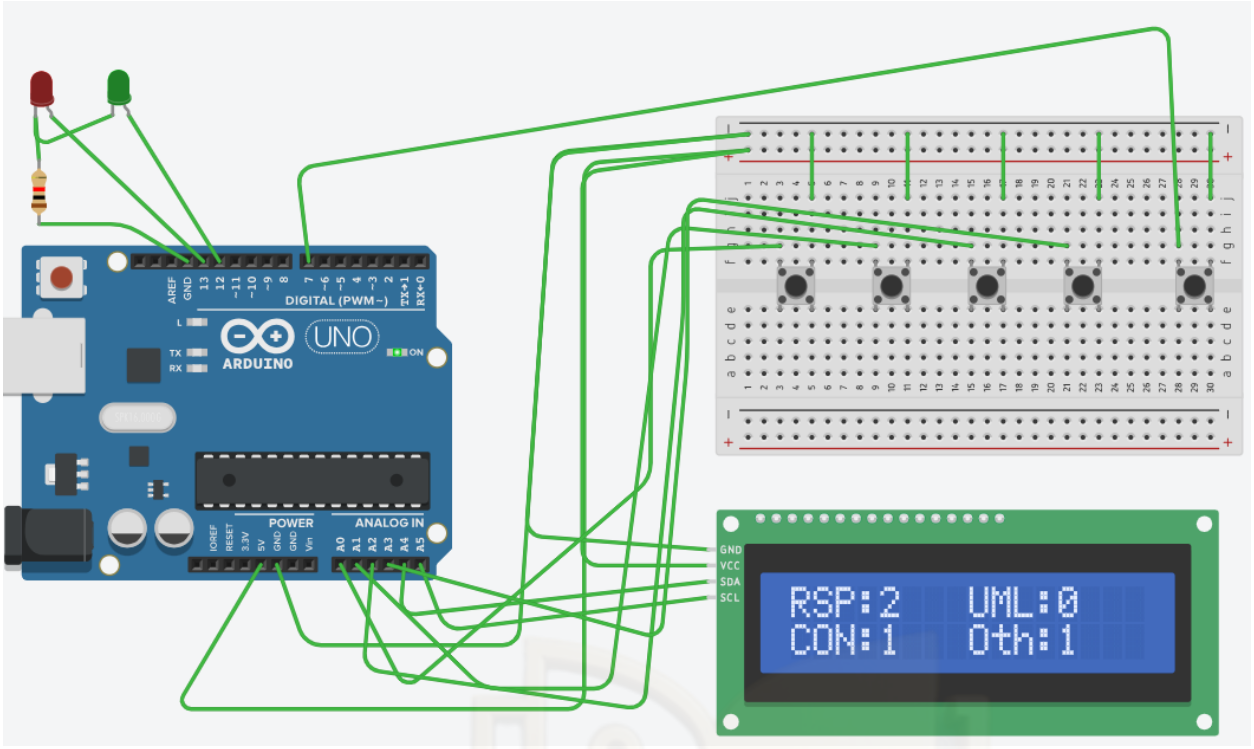
  if (key) {
    Serial.print("Key pressed: ");
    Serial.println(key);
    if (key == '#') {
      if (inputPassword == password) {
        Serial.println("Access Granted!");
        digitalWrite(ledPin, HIGH);
        noTone(buzzerPin);
      } else {
        Serial.println("Access Denied!");
        digitalWrite(ledPin, LOW);
        tone(buzzerPin, 1000, 500);
      }
      inputPassword = "";
    }
    else if (key == '*') {
      inputPassword = "";
      Serial.println("Password Cleared");
    }
    else {
      inputPassword += key;
    }
  }
}
```

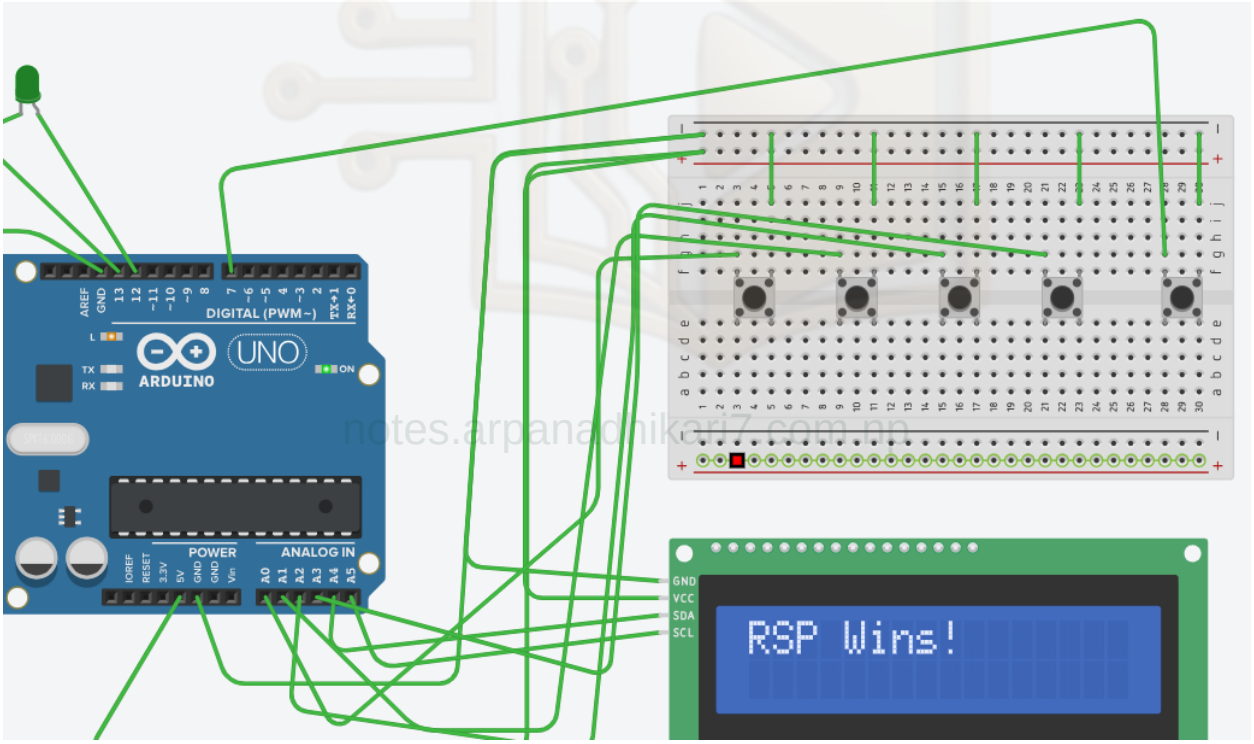
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18. Voting system with LCD:









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Code:

```
#include <Adafruit_LiquidCrystal.h>

Adafruit_LiquidCrystal lcd(0);
#define sw1 A0
#define sw2 A1
#define sw3 A2
#define sw4 A3
#define sw5 7   // Result button

int votes[4] = {0, 0, 0, 0};
void setup() {
  pinMode(sw1, INPUT_PULLUP);
  pinMode(sw2, INPUT_PULLUP);
  pinMode(sw3, INPUT_PULLUP);
  pinMode(sw4, INPUT_PULLUP);
  pinMode(sw5, INPUT_PULLUP);
  pinMode(12, OUTPUT); // Green LED
  pinMode(13, OUTPUT); // Red LED
  lcd.begin(16, 2);
  lcd.setCursor(0,0);
  lcd.print("   Voting System");
  lcd.setCursor(0,1);
  lcd.print("   Ready..");
  delay(2000);
  lcd.clear();
}
void loop() {
  lcd.setCursor(0,0); lcd.print("RSP:"); lcd.print(votes[0]);
  lcd.setCursor(8,0); lcd.print("UML:"); lcd.print(votes[1]);
  lcd.setCursor(0,1); lcd.print("CON:"); lcd.print(votes[2]);
  lcd.setCursor(8,1); lcd.print("Oth:"); lcd.print(votes[3]);
  if (digitalRead(sw1) == LOW) registerVote(0);
  if (digitalRead(sw2) == LOW) registerVote(1);
  if (digitalRead(sw3) == LOW) registerVote(2);
  if (digitalRead(sw4) == LOW) registerVote(3);
  if (digitalRead(sw5) == LOW) showResult();
}
void registerVote(int candidate) {
  votes[candidate]++;
  digitalWrite(12, HIGH); // Green LED ON
  delay(100);
  digitalWrite(12, LOW);
  while (digitalRead(sw1+candidate) == LOW); // Wait until released
}
```

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```ino

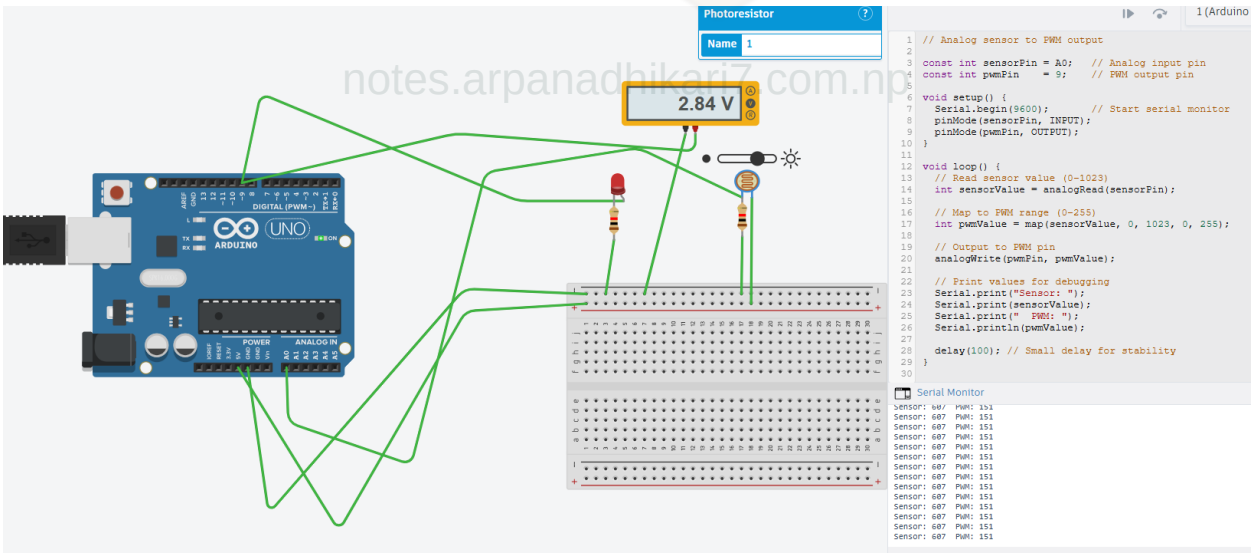
void showResult() {
  digitalWrite(13, HIGH); // Red LED ON

  int maxVotes = 0;
  int winner = -1;
  bool tie = false;

  for (int i = 0; i < 4; i++) {
    if (votes[i] > maxVotes) {
      maxVotes = votes[i];
      winner = i;
      tie = false;
    } else if (votes[i] == maxVotes && maxVotes > 0) {
      tie = true;
    }
  }
  lcd.clear();
  if (maxVotes == 0) {
    lcd.print("No Voting Yet");
  } else if (tie) {
    lcd.print("Tie / No Result");
  } else {
    const char* names[4] = {"RSP", "UML", "CON", "Oth"};
    lcd.print(names[winner]);
    lcd.print(" Wins!");
  }
  delay(5000);
  lcd.clear();
  for (int i = 0; i < 4; i++) votes[i] = 0;
  digitalWrite(12, LOW);
  digitalWrite(13, LOW);
}

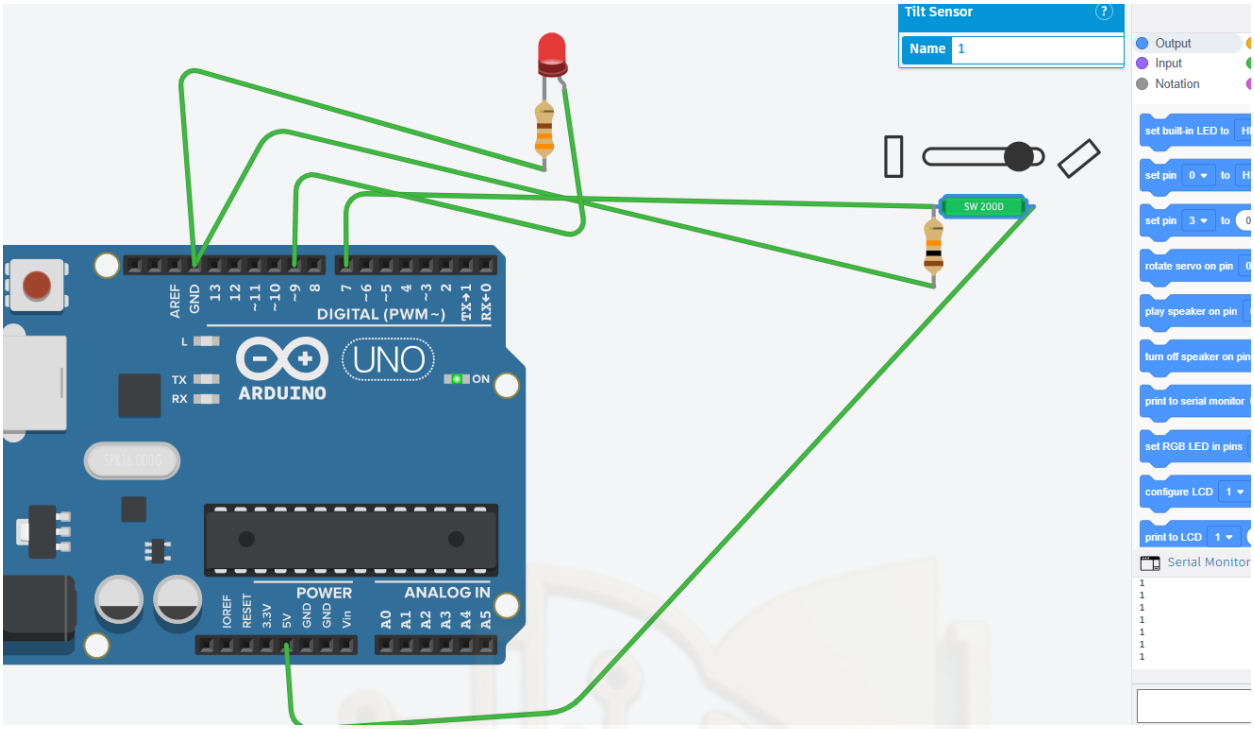
```

19. Light Intensity Measurement using LDR sensor: Circuit and code;

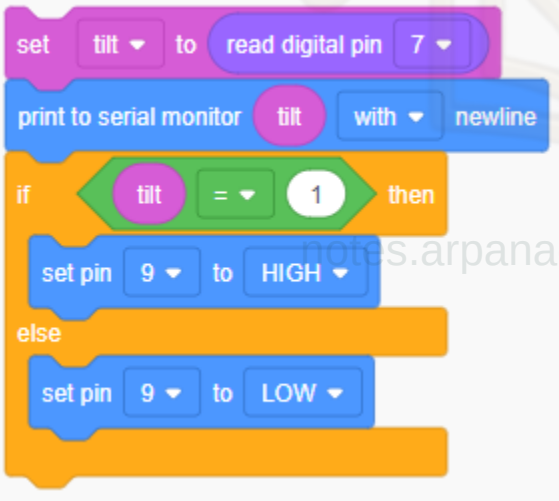


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20. Vibration sensor:



Code:



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